Booms, Shear Flow and Multi-Cell Torsion

Practice Guide

AERSP 301, HW 5 topics · a guide to work through with the video

How to use it. The video pauses at each checkpoint. Work it here, on paper or a tablet, before you press play. Checkpoints 1.1, 2.1 and 4.1 are answered by the video as soon as you resume. Checkpoints 3.1 and 5.1 are fresh practice problems that the video never solves: they repeat the homework's steps with new numbers, so check them against the full solutions on page 6. Stuck for five minutes? Read the hint on page 5.

Episode 1: Why Booms?

Checkpoint 1.1 A web in bending

video Ep 1, 2:11

A vertical web 300 mm deep and 10 mm thick is symmetric about the neutral axis of a beam in pure bending, and carries direct stress. When the section is idealized, how much area does this web add to the boom at each of its ends? Use $B_1 = \dfrac{t_D b}{6}\left(2 + \dfrac{\sigma_2}{\sigma_1}\right)$.

Your answer

Episode 2: Shear Flow With Booms

Checkpoint 2.1 Finish the channel

video Ep 2, 1:22

The channel has four booms of 300 mm^2: booms 1 and 2 at $y = +200$ mm (top flange), booms 3 and 4 at $y = -200$ mm (bottom flange), and $S_y/I_{xx} = 10^{-4}$ N/mm^4. Starting from $q = 0$ at the free edge, the flow in the top flange is $q_{12} = -6$ N/mm. Using $q_2 - q_1 = -\dfrac{S_y}{I_{xx}} B_r y_r$, find q_{23} (the web), q_{34} (the bottom flange) and the flow just past boom 4. What should the last one be, and why?

Your answer

Episode 3: Homework Problem 1

Checkpoint 3.1 **Practice box: booms, then the shear centre** video Ep 3, 3:24

A box section is 600 mm wide and 240 mm deep (centreline dimensions). Top and bottom skins are 8 mm thick, the left web 6 mm, the right web 5 mm. The left corners carry $50 \times 50 \times 6$ mm angles and the right corners $40 \times 30 \times 5$ mm angles. One material throughout. (a) Idealize it into four corner booms (skins and webs carry direct stress: use the ratio $+1$ for the skins and -1 for the webs in bending). (b) Find I_{xx}. (c) Number the booms as in the video (1 top left, 2 top right, 3 bottom right, 4 bottom left), cut the top wall, go $1 \to 2 \to 3 \to 4 \to 1$, and find the basic shear flow q_b in each wall for an upward load S_y. (d) Use zero twist to find $q_{s,0}$, then find the shear centre's distance ξ from the left web.

Your answer

Episode 4: Torsion of Multi-Cell Sections

Checkpoint 4.1 Twist rate of one cell

video Ep 4, 0:52

A two-cell box carries shear flows q_I and q_{II}, both taken counter-clockwise. Cell I has area A_I; the wall it shares with cell II has $\delta_{12} = \int ds/t$, and all of cell I's walls together (shared wall included) have δ_I. Write $d\theta/dz$ for cell I in terms of q_I, q_{II}, δ_I, δ_{12}, A_I and G.

Your answer

Episode 5: Homework Problem 2

Checkpoint 5.1 **Practice two-cell box: $T = 15$ kN m** video Ep 5, 2:07

A two-cell box is 250 mm deep. Cell I (on the left, with the left web) is 500 mm wide and cell II is 600 mm wide (rectangular cells, centreline dimensions; the spar between them is shared). Walls: left web 2.0 mm, spar 2.5 mm, right web 1.6 mm, all with $G = 27\,000$ N/mm^2; cell I's top and bottom skins 1.0 mm and cell II's top and bottom skins 0.8 mm, all with $G = 24\,000$ N/mm^2. It carries a counter-clockwise torque $T = 15$ kN m. Using $G_{REF} = 27\,000$ N/mm^2, find (a) t^* and δ for each wall, (b) q_I, q_{II} and the spar's shear flow, (c) the shear stress in each wall, $\tau = q/t$ with the real thickness, and (d) the rate of twist from $G_{REF}\, d\theta/dz = \frac{1}{2A_I}(q_I\delta_I - q_{II}\delta_{spar})$, in degrees per metre.

Your answer

Hints

Checkpoint 1.1 In pure bending, what are the stresses at the top and bottom of a web that is symmetric about the neutral axis? That gives σ_2/σ_1.

Checkpoint 2.1 Boom 3 is below the axis: its y_r is negative. A free edge has nothing to pass shear flow on to.

Checkpoint 3.1 An angle's area is (leg + leg − thickness) × thickness. Only the two webs have a basic flow; the cut wall and the opposite skin come out zero. For ξ, the web forces must add up to S_y.

Checkpoint 4.1 Go around cell I: which walls carry q_I, and which one carries $q_I - q_{II}$? Collect the q_I terms.

Checkpoint 5.1 Scale only the skins: $t^* = (G/G_{REF})t$. Set cell I's twist rate equal to cell II's, then use $T = 2A_I q_I + 2A_{II} q_{II}$ (with T in N mm).

Solutions

Checkpoint 1.1 A web in bending

In pure bending the web's two ends sit equal distances above and below the neutral axis, so their stresses are equal and opposite: $\sigma_2/\sigma_1 = -1$.

$$B = \frac{10 \times 300}{6}(2-1) = 500 \text{ mm}^2 \text{ to each boom.}$$

Check: a skin with the same stress at both edges ($\sigma_2/\sigma_1 = +1$) would give $t_D b/2 = 1500$ mm^2, three times as much; the web gives less because half of it is in tension and half in compression.

Checkpoint 2.1 Finish the channel

Each boom changes the flow by $-10^{-4} \times 300 \times y_r$:

$$\begin{aligned} q_{23} &= -6 - 10^{-4}(300)(200) = -6 - 6 = -12 \text{ N/mm} \\ q_{34} &= -12 - 10^{-4}(300)(-200) = -12 + 6 = -6 \text{ N/mm} \\ \text{past boom 4} &= -6 - 10^{-4}(300)(-200) = -6 + 6 = 0 \end{aligned}$$

The last one must be zero: the edge beyond boom 4 is free, so it carries no shear. Getting zero there is the check that every step was right. The minus signs mean the flow runs opposite to the direction s.

Checkpoint 3.1 Practice box: booms, then the shear centre

(a) Angles: left $(50+50-6)\times 6 = 564$ mm^2, right $(40+30-5)\times 5 = 325$ mm^2. Skins (ratio $+1$): $\dfrac{8\times 600}{6}(2+1) = 2400$ mm^2 to each corner. Webs (ratio -1): left $\dfrac{6\times 240}{6}(1) = 240$, right $\dfrac{5\times 240}{6}(1) = 200$ mm^2.

$$B_{left} = 564 + 2400 + 240 = 3204 \text{ mm}^2, \qquad B_{right} = 325 + 2400 + 200 = 2925 \text{ mm}^2.$$

(b) Symmetric about the horizontal axis, so $I_{xy} = 0$ and $I_{xx} = 2(3204+2925)(120)^2 = 176.5\times 10^6$ mm^4.

(c) Booms 1 (top left), 2 (top right), 3 (bottom right), 4 (bottom left); cut wall 12 and go $1 \to 2 \to 3 \to 4 \to 1$:

$$\begin{aligned} q_{b,12} &= 0 \\ q_{b,23} &= -\frac{S_y}{I_{xx}}(2925)(120) = -1.989 \times 10^{-3}\, S_y \\ q_{b,34} &= q_{b,23} - \frac{S_y}{I_{xx}}(2925)(-120) = 0 \\ q_{b,41} &= 0 - \frac{S_y}{I_{xx}}(3204)(-120) = +2.178 \times 10^{-3}\, S_y \end{aligned}$$

(d) At the shear centre the section does not twist: $\oint (q_b + q_{s,0})\, ds/t = 0$, with $ds/t =$ top $600/8 = 75$, right $240/5 = 48$, bottom 75, left $240/6 = 40$ (total 238):

$$\begin{aligned} &(-1.989 \times 10^{-3})(48) + (2.178 \times 10^{-3})(40) + 238\, q_{s,0} = 0 \\ &-95.45 \times 10^{-3} + 87.13 \times 10^{-3} + 238\, q_{s,0} = 0 \;\Rightarrow\; q_{s,0} = 3.50 \times 10^{-5}\, S_y. \end{aligned}$$

Totals: $q_{23} = -1.954 \times 10^{-3} S_y$ and $q_{41} = 2.213 \times 10^{-3} S_y$ (both webs flow upward). Web forces: $1.954 \times 10^{-3} \times 240 = 0.469\, S_y$ and $2.213 \times 10^{-3} \times 240 = 0.531\, S_y$; together $1.000\, S_y$. ✓

Moments about the left web: the right web gives $0.469\, S_y \times 600 = 281.3\, S_y$; the skins' small flows form an opposite couple $3.50 \times 10^{-5} S_y \times 600 \times 240 = 5.0\, S_y$. So

$$S_y\, \xi = 281.3\, S_y - 5.0\, S_y \;\Rightarrow\; \xi = 276.3 \text{ mm from the left web,}$$

left of the middle (300 mm) because the left booms are heavier, as in the homework.

Checkpoint 4.1 Twist rate of one cell

Around cell I every wall carries q_I except the shared wall, which carries $q_I - q_{II}$:

$$\begin{aligned}\frac{d\theta}{dz} &= \frac{1}{2A_I G} \oint q\, \frac{ds}{t} = \frac{1}{2A_I G}\Big[q_I(\delta_I - \delta_{12}) + (q_I - q_{II})\delta_{12}\Big] \\ &= \frac{1}{2A_I G}\big(q_I \delta_I - q_{II}\delta_{12}\big).\end{aligned}$$

Checkpoint 5.1 Practice two-cell box: $T = 15$ kN m

(a) Webs and spar keep their thickness. Skins get $t^* = (24\,000/27\,000)\, t = 0.8889\, t$, so 0.889 mm (cell I) and 0.711 mm (cell II). Then $\delta = L/t^*$:

$$\begin{aligned}&\text{left web } 250/2.0 = 125, \quad \text{spar } 250/2.5 = 100, \quad \text{right web } 250/1.6 = 156.3, \\ &\text{each cell-I skin } 500/0.889 = 562.5, \quad \text{each cell-II skin } 600/0.711 = 843.8.\end{aligned}$$

$$\delta_I = 125 + 2(562.5) + 100 = 1350, \qquad \delta_{II} = 156.25 + 2(843.75) + 100 = 1943.75.$$

(b) Areas $A_I = 500 \times 250 = 125\,000$ mm^2, $A_{II} = 600 \times 250 = 150\,000$ mm^2. Equal twist:

$$\begin{aligned}&\frac{1350\, q_I - 100\, q_{II}}{2(125\,000)} = \frac{-100\, q_I + 1943.75\, q_{II}}{2(150\,000)} \\ &\Rightarrow\; 150\,000(1350 q_I - 100 q_{II}) = 125\,000(-100 q_I + 1943.75 q_{II}) \\ &\Rightarrow\; 215.0 \times 10^6\, q_I = 258.0 \times 10^6\, q_{II} \;\Rightarrow\; q_I = 1.200\, q_{II}.\end{aligned}$$

Torque, with $T = 15 \times 10^6$ N mm: $15 \times 10^6 = 2(125\,000)(1.200\, q_{II}) + 2(150\,000)\, q_{II} = 600\,000\, q_{II}$, so

$$q_{II} = 25.0 \text{ N/mm}, \qquad q_I = 30.0 \text{ N/mm}, \qquad q_{spar} = q_I - q_{II} = 5.0 \text{ N/mm}.$$

Both cells flow counter-clockwise, with the torque; in the spar the net flow follows cell I.

(c) $\tau = q/t$ with the real thickness: left web $30.0/2.0 = 15.0$; cell-I skins $30.0/1.0 = 30.0$; spar $5.0/2.5 = 2.0$; cell-II skins $25.0/0.8 = 31.3$; right web $25.0/1.6 = 15.6$ N/mm^2.

(d)

$$G_{REF}\, \frac{d\theta}{dz} = \frac{30.0(1350) - 25.0(100)}{2(125\,000)} = \frac{40\,500 - 2\,500}{250\,000} = 0.152,$$

so $\dfrac{d\theta}{dz} = \dfrac{0.152}{27\,000} = 5.63 \times 10^{-6}$ rad/mm $= 0.32°$/m. Check with cell II: $\dfrac{-30.0(100) + 25.0(1943.75)}{2(150\,000)} = \dfrac{45\,594}{300\,000} = 0.152$. ✓